

Set Theory

Outline

- Sets-

- Describing sets, notation, examples

- Set equality

- Membership relationships

- Set Operations-

 - Union, Intersection, Complement and Difference

- Venn Diagrams

- DeMorgan's Laws for sets

- Set partition

- Cardinality, Power set ,Relations, Cartesian product, Functions

Set Theory

- Set: A *Set* is any well-defined collection of “objects.”
- $A = \{a_1, a_2, a_3, \dots, a_n\}$ “A contains...”
- Denote sets with upper-case letters, elements with lower-case letters.

The following notation is used to show set membership

- $a_1 \in A$ “ a_1 is an element of A”
 “ a_1 is a member of A”
- $b_1 \notin A$ “ b_1 is not an element of A”

Set Theory

- If B is the set of letters in the English Alphabet, then we have,
 $B = \{a, b, c, \dots, x, y, z\}$
- We can write $a \in B, b \in B, c \in B, \dots, z \in B$
- $n(B) = 26$
- Note: **Number of elements of a given set A is denoted by $n(A)$.** In the above example since the number of letters in the Alphabet is 26, $n(B)=26$.

Ways of Describing Sets

- List the elements

$$A = \{1, 2, 3, 4, 5, 6\}$$

- Give a verbal description
 - “A is the set of all positive integers from 1 to 6, inclusive”

- Denote using the set builder notation / mathematical notation

$$A = \{x \mid x \in \mathbb{Z}^+, x < 7\}$$

- Read as
 - “A is the set of x such that x is a positive integer and less than 7”

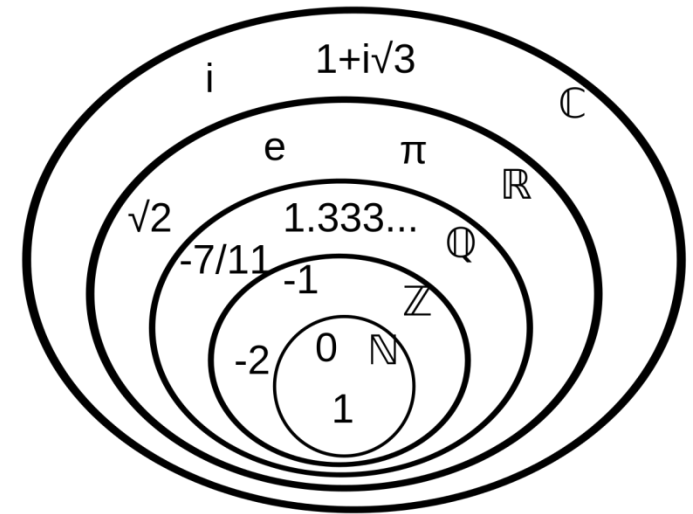
Examples for Sets

“Standard” Sets:

- Natural numbers $\mathbb{N} = \{1, 2, 3, \dots\}$
- Integers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$
- Positive Integers $\mathbb{Z}^+ = \{1, 2, 3, 4, \dots\}$
- Real Numbers $\mathbb{R} = \{47.3, -12, \pi, \dots\}$
- Rational Numbers $\mathbb{Q} = \{1.5, 2.6, -3.8, 15, \dots\}$

$\mathbb{Q}$ denotes the set of rational numbers, then

$$\mathbb{Q} = \{x \mid x = p/q \text{ where } p \in \mathbb{Z}, q \in \mathbb{Z}, q \neq 0\}$$



Set Equality

- Sets A and B are equal if and only if they contain exactly the same elements.

Example

- $A = \{9, 2, 7, -3\}$, $B = \{7, 9, -3, 2\}$:

$$A = B$$

Order of elements is meaningless

Q) Identify which sets are equal

$$A = \{x \mid x \text{ is an integer where } x > 0 \text{ and } x < 5\}$$

$$B = \{x \mid x \text{ is a positive integer whose square is } > 0 \text{ and } < 25\}$$

$$C = \{x \mid x \text{ is an integer whose square is } > 0 \text{ and } < 25\}$$

Some Special Sets

- The Null Set or Empty Set $\{ \}$. This is a set with no elements, often symbolized by



- The Universal Set. This is the set of all elements currently under consideration, and is often symbolized by

Ω

U

- Also symbolized by

Membership Relationships

Subset.

$$A \subseteq B \quad \text{"A is a subset of B"}$$

We say "A is a subset of B" if $x \in A \Rightarrow x \in B$,

i.e., all the members of A are also members of B. The notation for subset means, in terms of the sets, "included in or equal to."

- $B = \{9, 2, 7, -3\}$,
 $\{9\}$, $\{9, 2\}$, $\{2, 7, -3\}$ are subsets of B and $\{9, 2, 7, -3\}$ is also a subset of B

Membership Relationships

Proper Subset

$$A \subset B$$

“A is a proper subset of B”

We say “A is a proper subset of B” if all the members of A are also members of B, but in addition there exists at least one element c such that $c \in B$ but $c \notin A$.

The notation for proper subset means, in terms of the sets, “included in but not equal to.”

- $\{9, 2, 7, -3\}$ is a subset of the set $\{9, 2, 7, -3\}$ but not a proper subset of the set $\{9, 2, 7, -3\}$.

Questions

Fill in the blanks with $\in$, $\notin$, $\subseteq$, $=$ or $\neq$

Recall that $\mathbb{Z}$ is the set of all integers and $\emptyset$ is the empty set, $\{\}$.

- | | | | |
|----|------------------|-------|------------------------|
| a. | 2 | ----- | $\{2, 4, 6\}$ |
| b. | $\{2\}$ | ----- | $\{2, 4, 6\}$ |
| c. | 1.5 | ----- | $\mathbb{Z}$ |
| d. | -1.5 | ----- | $\mathbb{Z}$ |
| e. | 15 | ----- | $\mathbb{Z}$ |
| f. | $\{-15\}$ | ----- | $\mathbb{Z}$ |
| g. | $\emptyset$ | ----- | $\mathbb{Z}$ |
| h. | 54 | ----- | $\{6, 12, 18, \dots\}$ |
| i. | 54 | ----- | $\{6, 12, 18\}$ |
| j. | $\{1, 3, 3, 5\}$ | ----- | $\{1, 3, 5\}$ |
| k. | $\{-3, 1, 5\}$ | ----- | $\{1, 3, 5\}$ |
| l. | $\{3, 1, 5\}$ | ----- | $\{1, 3, 5\}$ |

Combining Sets – Set Union

$$A \cup B$$

- “A union B” is the set of all elements that are in A, or B, or both.

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

$$A = \{1, 2, 3\}$$

$$B = \{3, 4, 5, 6\}$$

$$A \cup B = \{1, 2, 3, 4, 5, 6\}$$

Combining Sets – Set Intersection

$$A \cap B$$

- “A intersect B” is the set of all elements that is in both A and B.

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

$$A = \{1, 2, 3\}$$

$$B = \{3, 4, 5, 6\}$$

$$A \cap B = \{3\}$$

Set Complement A'

This is also denoted by $\overline{A}$

- “A complement,” or “not A” is the set of all elements not in A.

$$\overline{\overline{A}} = A$$

$$\Omega = \{1, 2, 3, 4, 5, 6\} \quad B = \{3, 4, 5, 6\}$$

$$\overline{B} = \{1, 2\}$$

Set Difference

$$A - B$$

- The set difference “A minus B” is the set of elements that are in A, with those that are in B subtracted out. Another way of putting it is, it is the set of elements that are in A, and not in B, so

$$A - B = A \cap \bar{B}$$

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

$$A = \{3, 4, 5, 6\}$$

$$B = \{1, 2, 3\}$$

$$A - B = \{4, 5, 6\}$$

Questions

Answer the following questions using the information given.

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$A = \{2, 4, 6, 8, 10\}$$

$$B = \{1, 3, 7\}$$

$$C = \{3, 7\}$$

Required:

a) Subsets of C

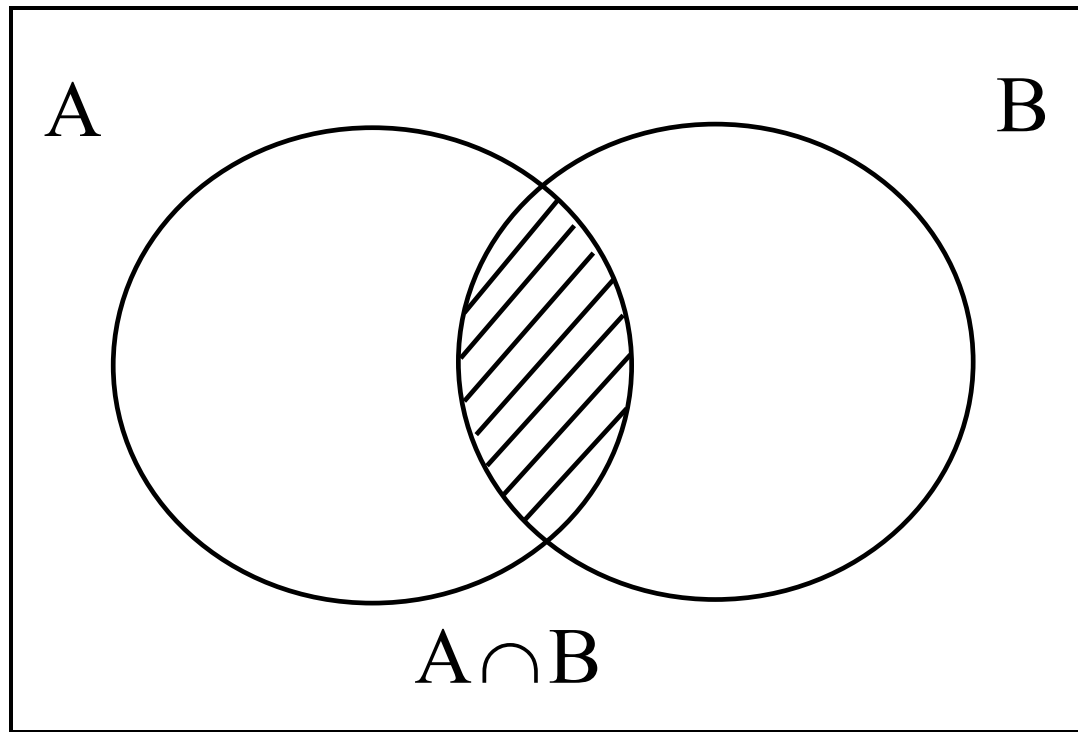
b) Proper subsets of B

c) $A \cap B$, $A \cup C$, A' , B' ,

$B \cap A'$, $B \cap C'$, $A - B$

Venn Diagrams

- Venn Diagrams use topological areas to stand for sets. I've done this one for you.



Exercise - Venn Diagrams

Identify the regions in separate Venn diagrams for following:

- a. $A \cup B$
- b. A' (also symbolized by $\overline{A}$)
- c. B' (also symbolized by $\overline{B}$)
- d. $A - B$

Venn Diagrams

- Shade the identified region in the following expressions

$$\overline{A} \cap \overline{B}$$

$$\overline{A \cup B}$$

$$\overline{A \cap B}$$

$$\overline{A} \cup \overline{B}$$

De Morgan's Law for Sets

$$\overline{A \cup B} = \bar{A} \cap \bar{B}$$

$$\overline{A \cap B} = \bar{A} \cup \bar{B}$$

Let's use Venn Diagrams to verify DeMorgan's law

De Morgan's Law for Sets

$$\overline{(A \cup B) \cap C} = (\overline{A} \cap \overline{B}) \cup \overline{C}$$

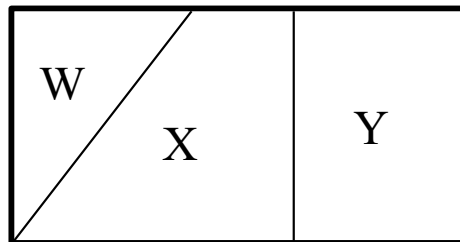
Mutually Exclusive and Exhaustive Sets

- Definition. We say that a group of sets is exhaustive of another set if their union is equal to that set. For example, if $A \cup B = C$ we say that A and B are exhaustive with respect to C.
- Definition. We say that two sets A and B are mutually exclusive if $A \cap B = \emptyset$, that is, the sets have no elements in common.

Set Partition

- Definition. We say that a group of sets partitions another set if they are mutually exclusive and exhaustive with respect to that set. When we “partition a set,” we break it down into mutually exclusive and exhaustive regions, i.e., regions with no overlap. The Venn diagram below should help you get the picture. In this diagram, the set A (the rectangle) is partitioned into sets W, X, and Y.

A



Some Test Questions

$$A \cup \emptyset = ?$$

Some Test Questions

$$A \cup \bar{A} = ?$$

Some Test Questions

$$A \cap \bar{A} = ?$$

Some Test Questions

$$A \cup \Omega = ?$$

Some Test Questions

$$A \cap \Omega = ?$$

Some Test Questions

If $A \subset B$ then

$$A \cap B = ?$$

Some Test Questions

If $A \subset B$ then
 $A \cup B = ?$